

	Work Instruction	Doc. Revision	01
	AXI V810i AKD Kollmorgen Drive Firmware and Parameter Upgrade Procedure	Effective Date	12 Dec 2018

This procedure is applicable to below models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL		√	
V810 / V810i S2EX		√	
V810 / V810i Mini			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	12 Dec 2018	First Release	Chin Chee Yan
01	20 Dec 2018	Figure Update	Ong Kean Sheng

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X Axis AKD Kollmorgen Drive Upgrade Procedure

1. Go to Connected Processor tab with V810 GUI, click the "Stop RMS" # **MUST** perform

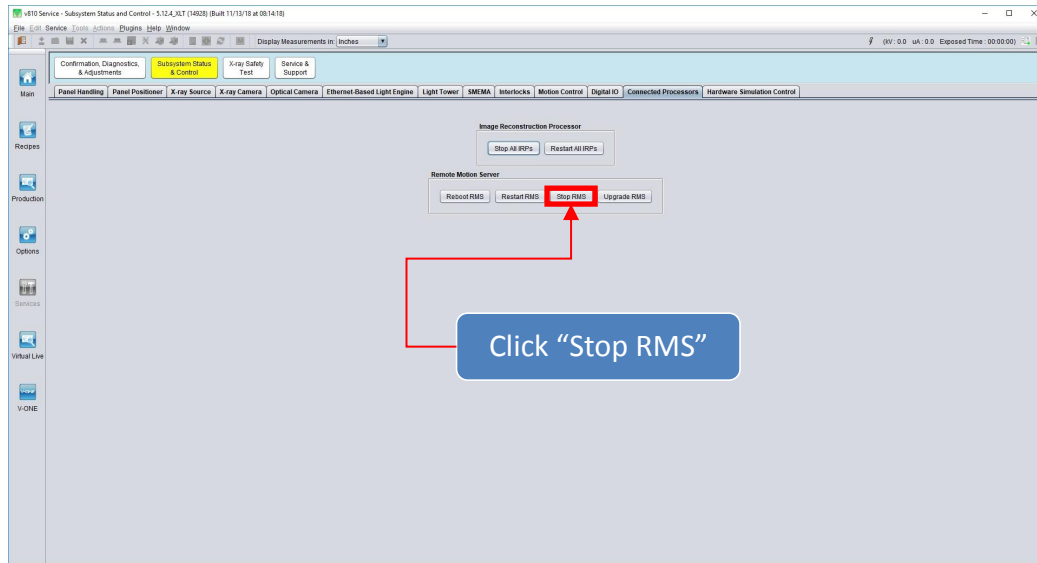


Figure 1 Stop RMS Driver

2. Prepare a LAN (CAT5) cable with 1.5 meter length.



Figure 2 LAN Cable

3. Open the server rack cover & MCA/SSCA Box cover.
4. Connect the LAN (CAT5) cable to the Atom PC on board LAN port (origin empty port).

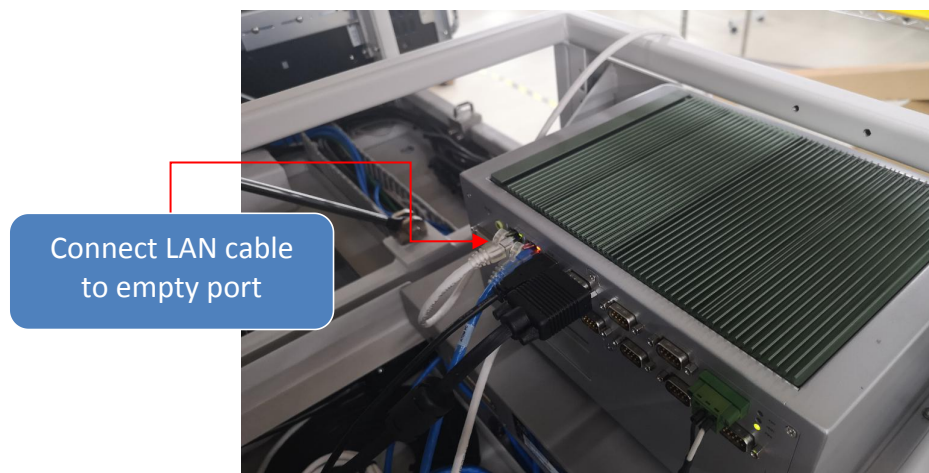


Figure 3 Connect LAN Cable to Atom PC on board LAN port

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5. Connect the other end of the LAN (CAT5) cable to the X-Axis AKD Kollmorgen Drive.

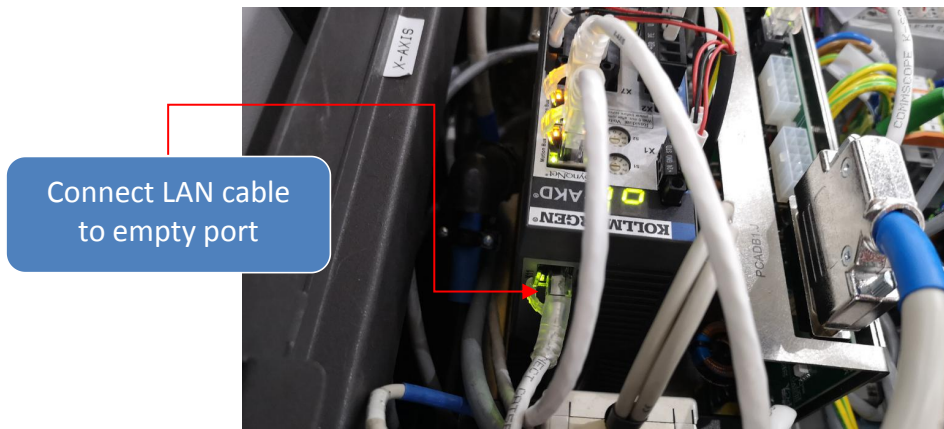


Figure 4 Connect LAN Cable to X-Axis AKD Kollmorgen Drive

6. Power cycle the SSCA/MCA Box power with toggle the switch at the side panel. # **MUST** perform

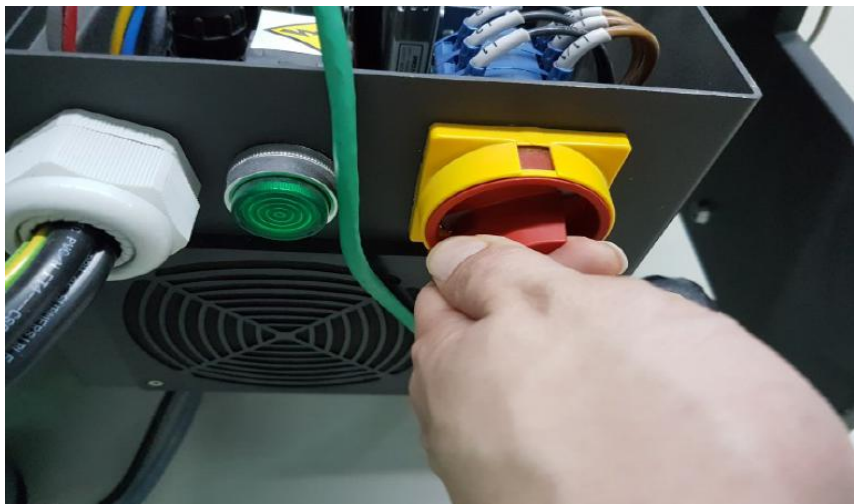


Figure 5

7. At main server, launch the “Remote Desktop Connection” application.

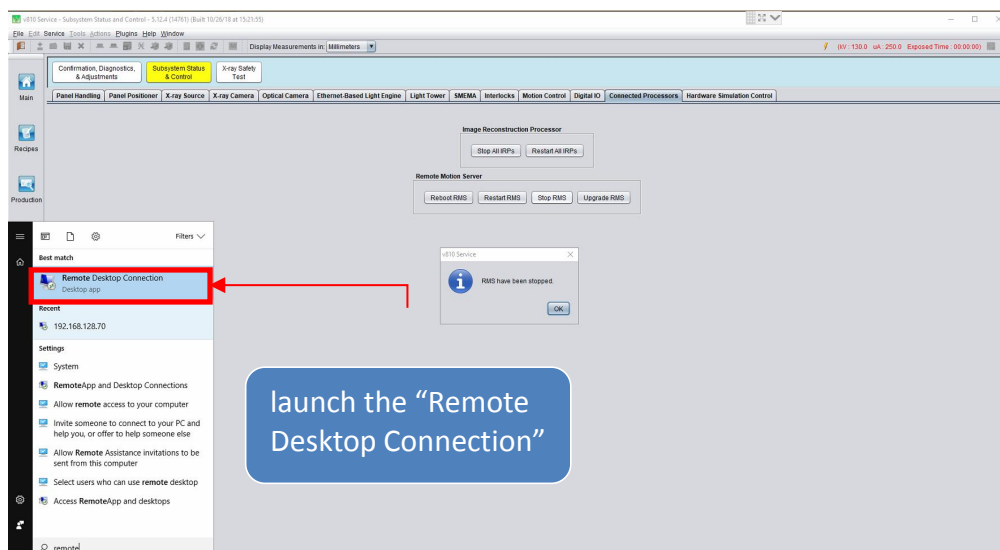


Figure 6: Launch “Remote Desktop Connection” application

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8. Remote desktop to Atom PC with IP Address **192.168.128.70**, Username, **Administrator**, Password, **Please!**

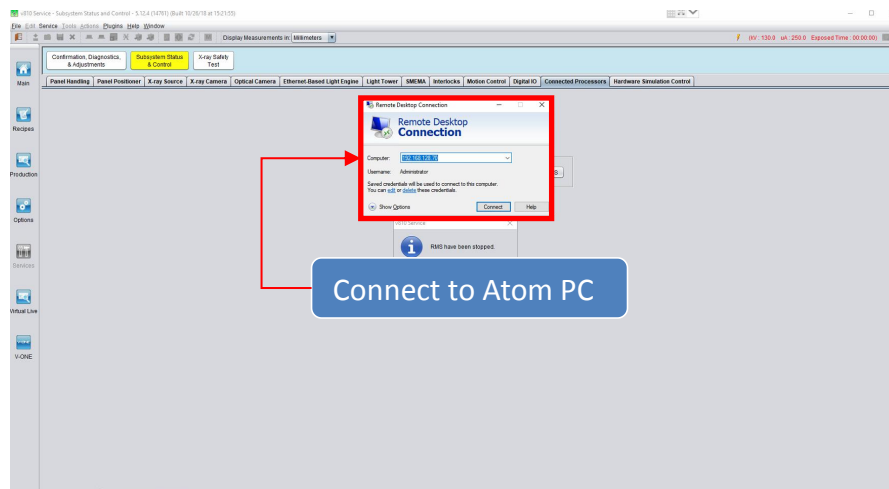


Figure 7: Launch “Remote Desktop Connection” application

9. Launch the “Kollmorgen WorkBench” application with clicking the Icon at Desktop. Click the “New Project”. (Remark, the driver pre-installation would be require)

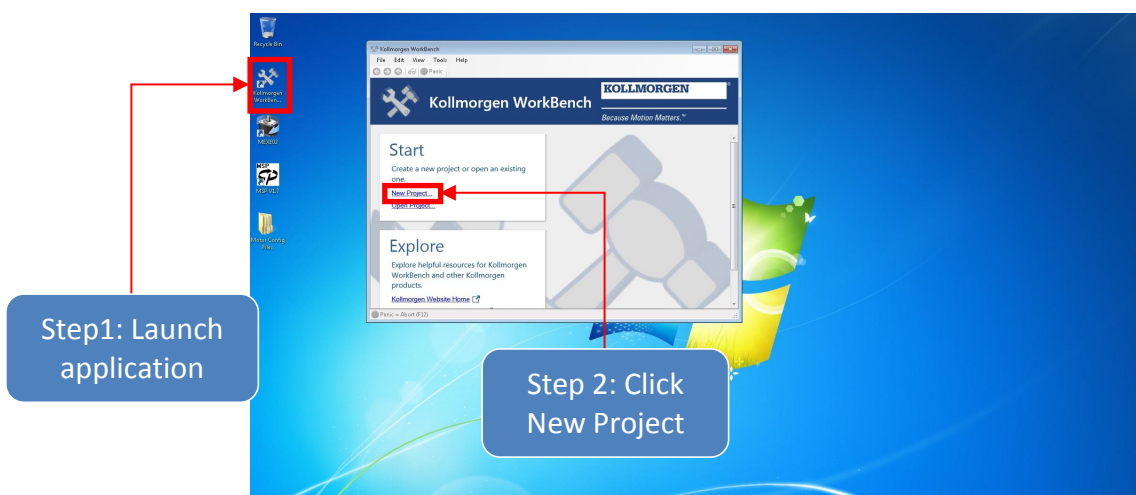


Figure 8: Launch “Kollmorgen WorkBench” application and select “New Project”

10. The interface below will shown, select the kollmorgen drive listed and click “Connect”.

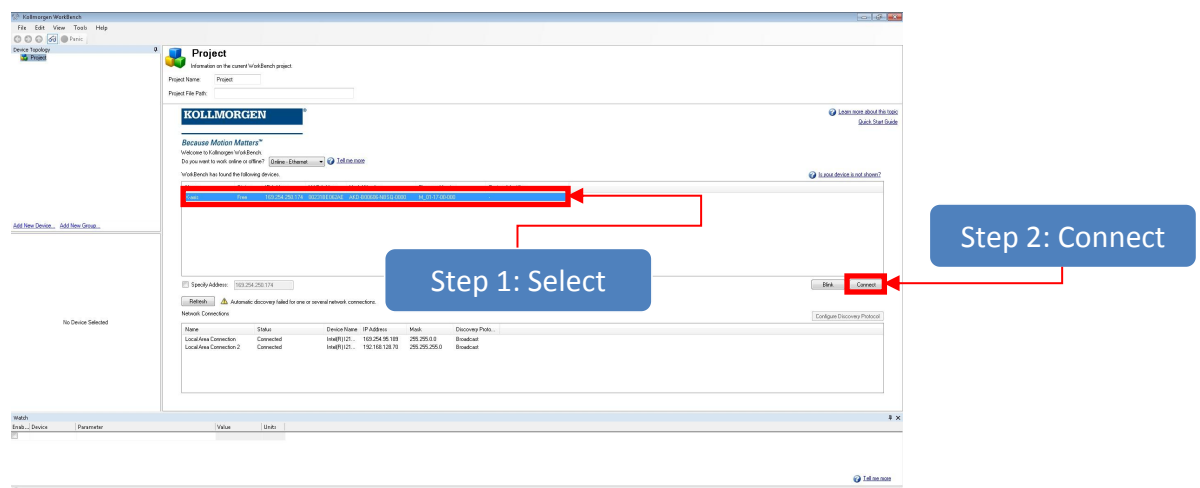


Figure 9: Log in the connected drive

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Upgrade X-Axis Kollmorgen Drive Firmware Update

11. The interface below will shown, Rename the drive with “X-Axis”click “More” continues with “Update”

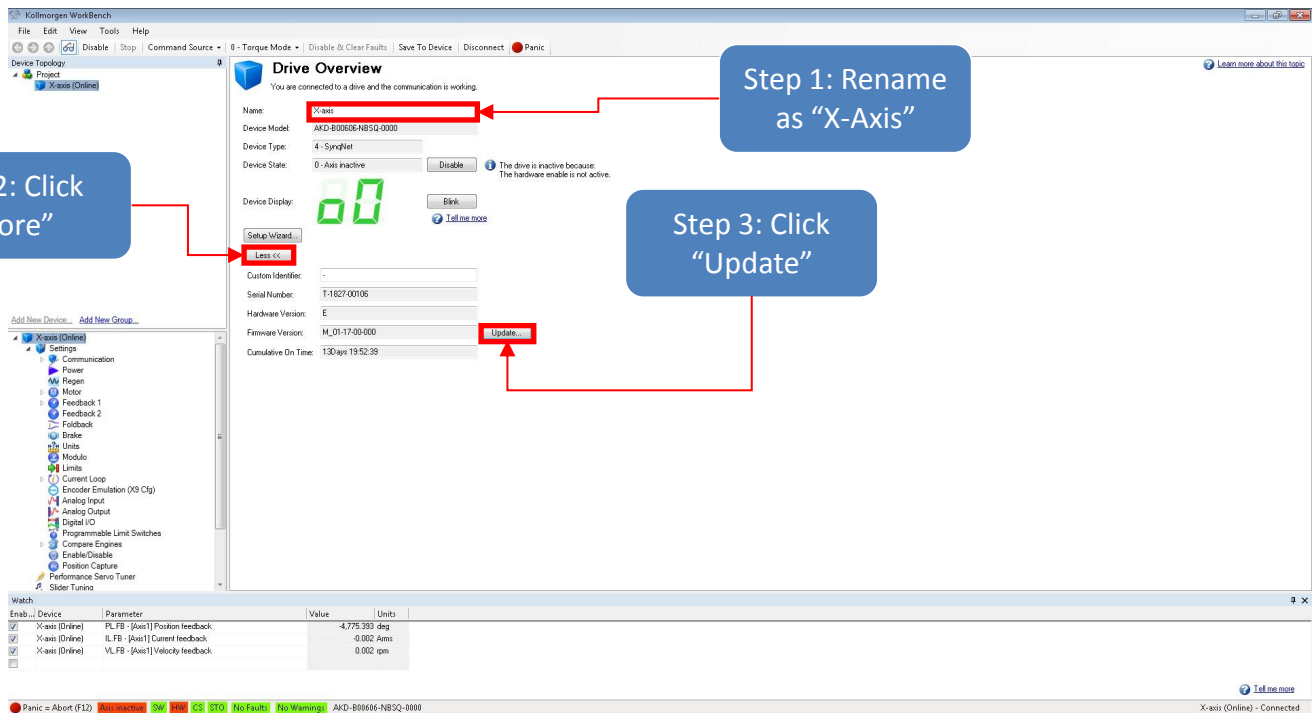


Figure 10: Log in the connected drive

12. Software will remind user that driver name has been changed and ask user whether to save this to device or not. Select “Yes” to continue

13. Go to path “C:\Program Files\AXI System\v810\5.0<Config>\motionControl\motionDrive\synqNet\AXI AKD Tuning Files” and select the “AKD-B-NBSQ-V01-17-00-000.i00”. Click “Open”.

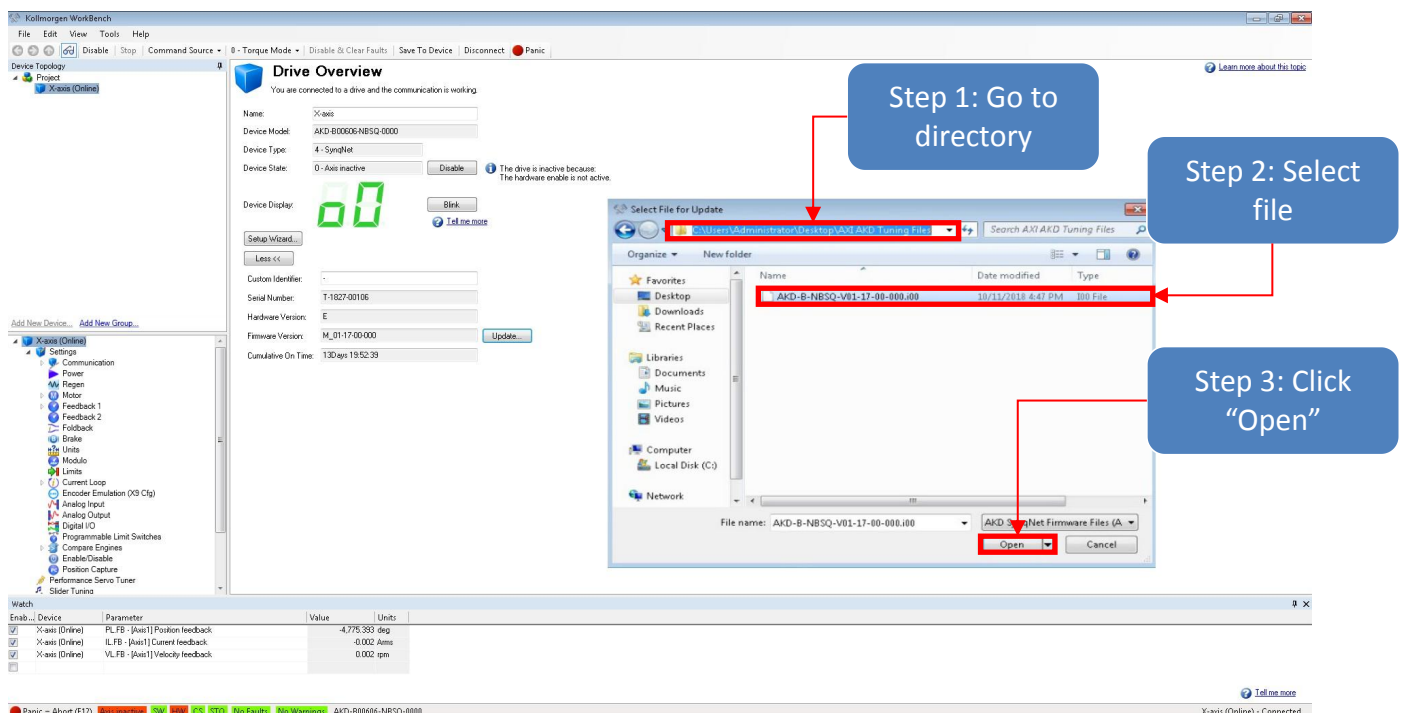


Figure 11: Select Firmware file

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14. The software will appear as loading status. It will takes few minutes to load, please wait until it finish and the driver status panel will appear again. This means the firmware is loaded successfully.

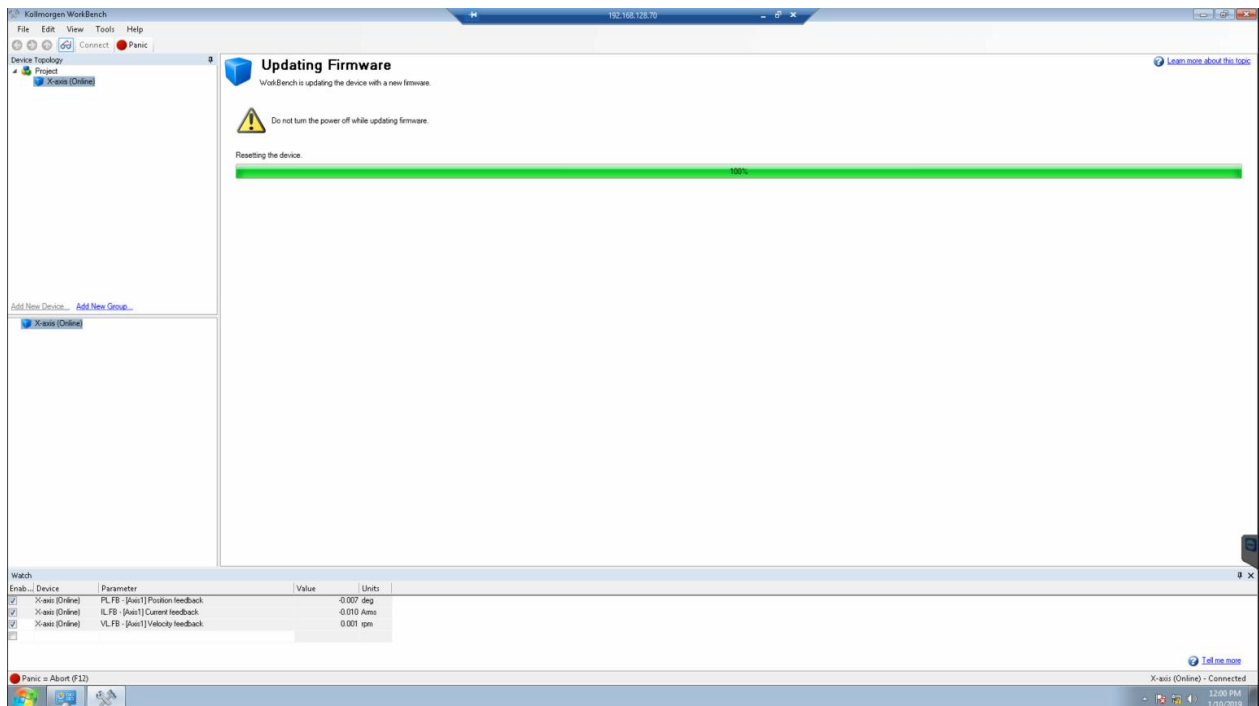


Figure 12: Complete update Firmware

Upgrade X-Axis Kollmorgen Drive Parameter Update

15. At the same Kollmorgen WorkBench” application, click the “Tools” > “X Axis Online” > “Load Parameter”.

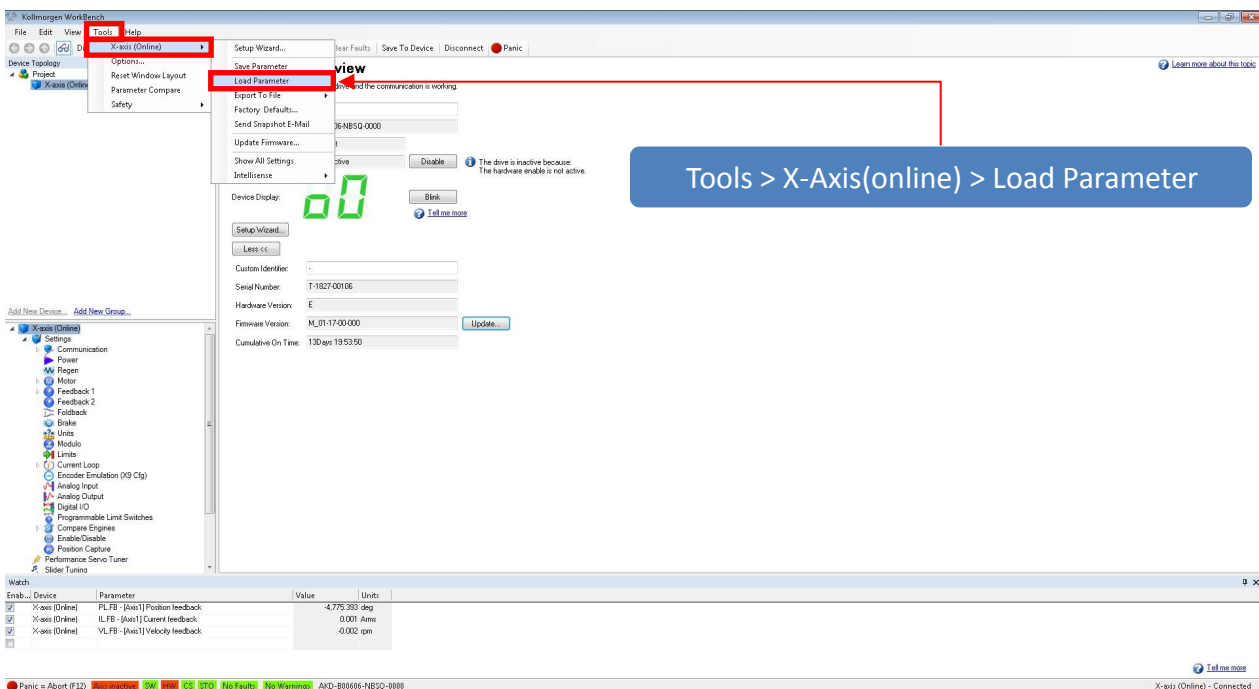


Figure 13: go to “Load Parameter” function

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16. Go to path “C:\Program Files\AXI System\v810\5.0\<Config>\motionControl\motionDrive\synqNet\AXI AKD Tuning Files” and select the “x_axis.akd”. Click “Open”.

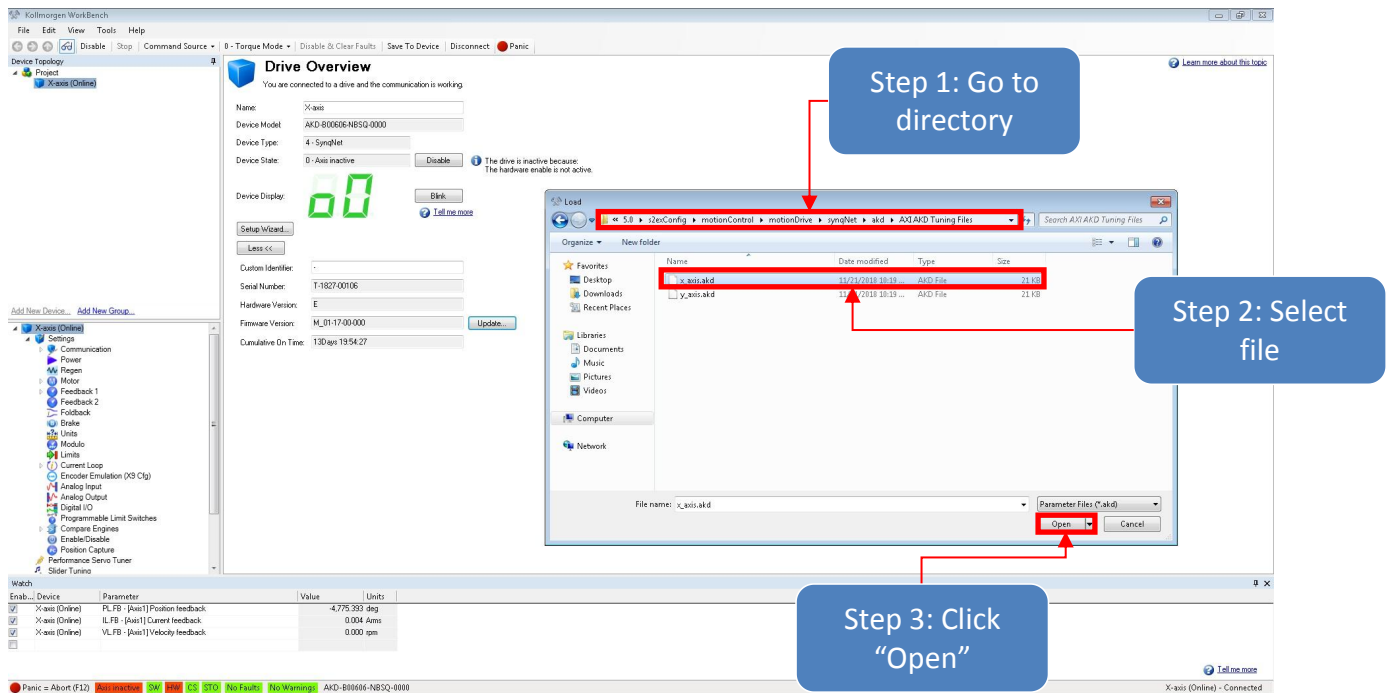


Figure 14: Select X Axis Parameter file

17. A loading dialog will prompt out. When the parameter loaded successfully, a smiley face will show up. Then click “Save To Device” button and follow with “OK”.

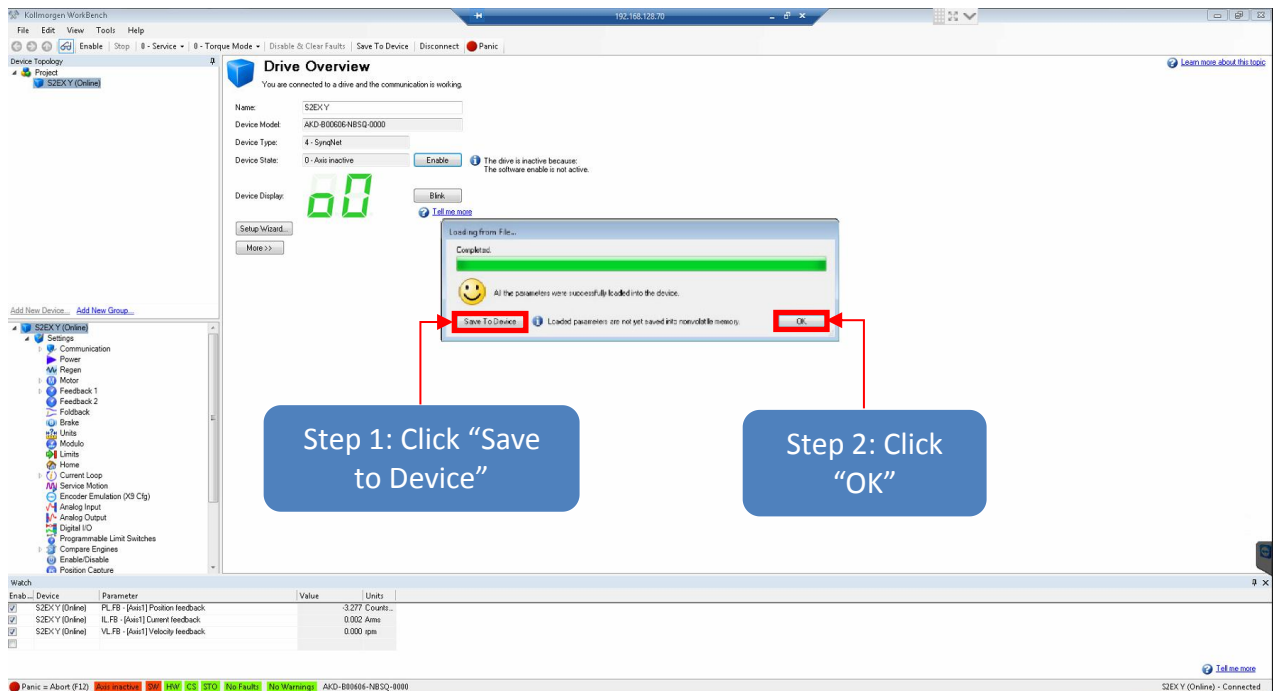


Figure 15: Save Parameter setting to X axis driver

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18. Once complete, exit the “Kollmorgen WorkBench” and click “No”. Close all the window.

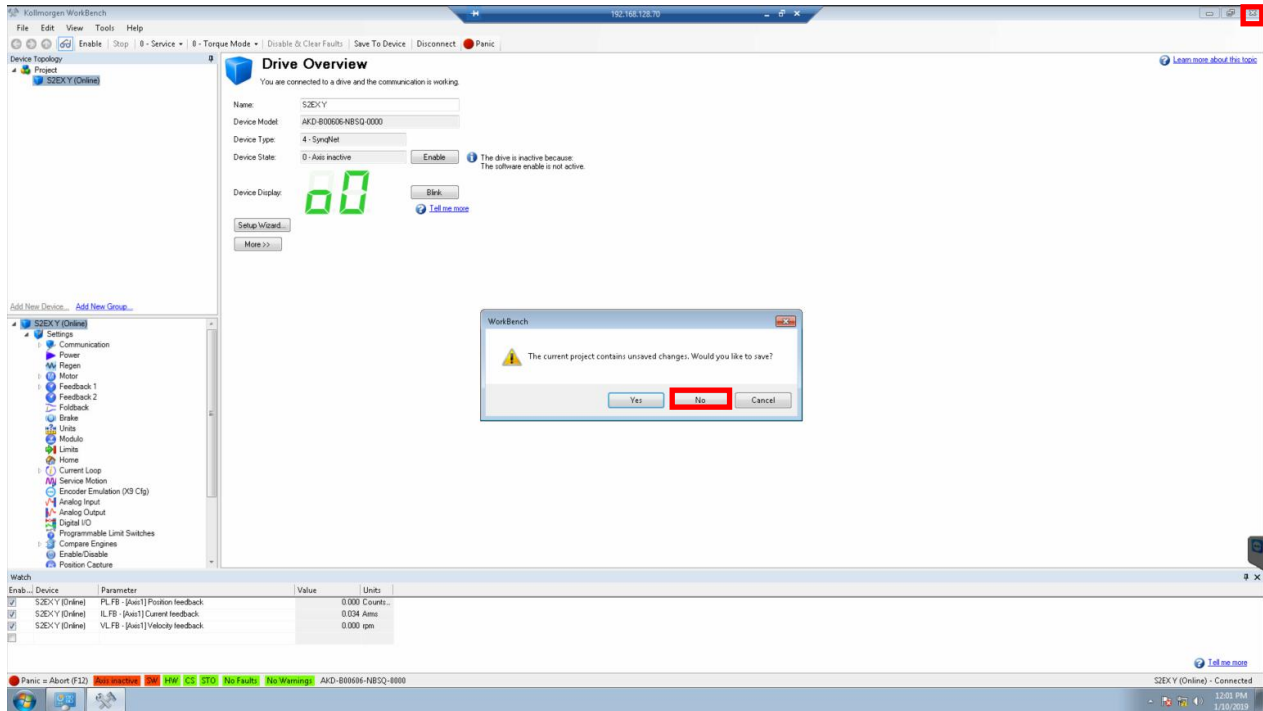


Figure 16: Save Parameter setting to X axis driver

19. Back to V810 GUI, Connected processor, click Stop > Upgrade > Restart RMS and next to Motion Control Tab and click “ Upgrade Firmware”.

Y Axis AKD Kollmorgen Drive Upgrade Procedure

20. May refer to X Axis AKD Upgrade Step step 1 to 8 while replace step 5 with step 21 (Connect external LAN to Y-Axis AKD Kollmorgen Drive).

21. Connect the other end of the LAN (CAT5) cable to the Y-Axis AKD Kollmorgen Drive.

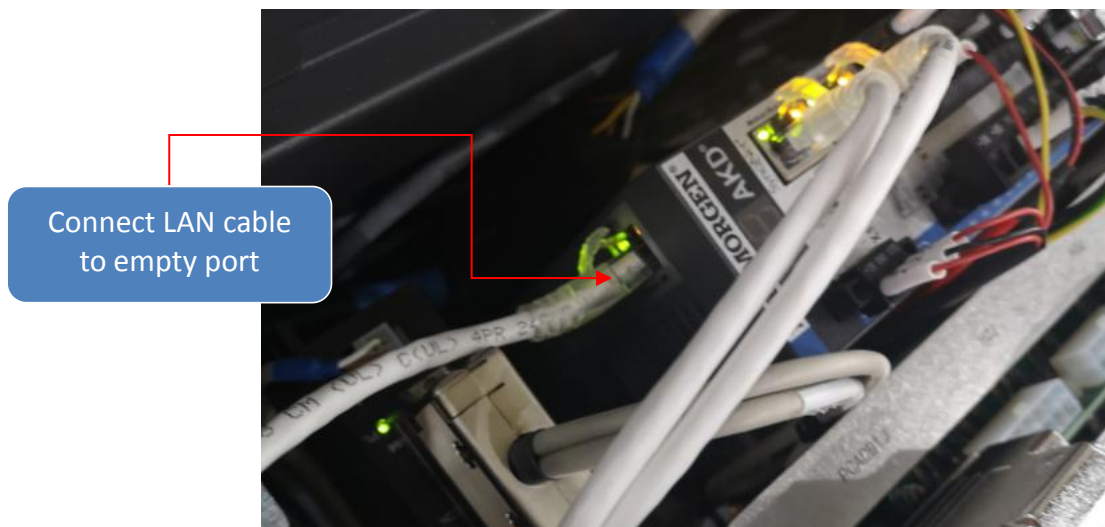


Figure 17: Connect LAN Cable to Y-Axis AKD Kollmorgen Drive

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22. Launch the “Kollmorgen WorkBench” application with clicking the Icon at Desktop. Click the “New Project”.
- (Remark, the driver pre-installation would be require)

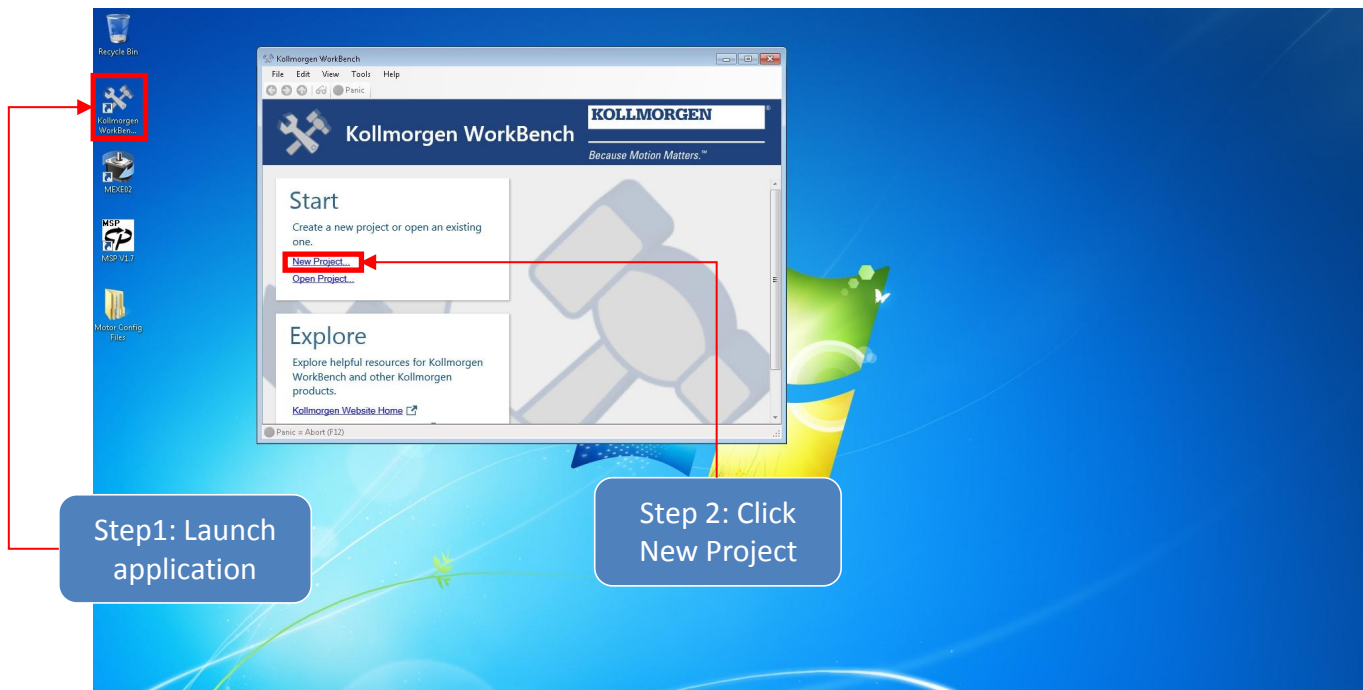


Figure 18: Launch “Kollmorgen WorkBench” application and select “New Project”

23. The interface will illustrate as below, select the kollmorgen drive listed and click “Connect”.

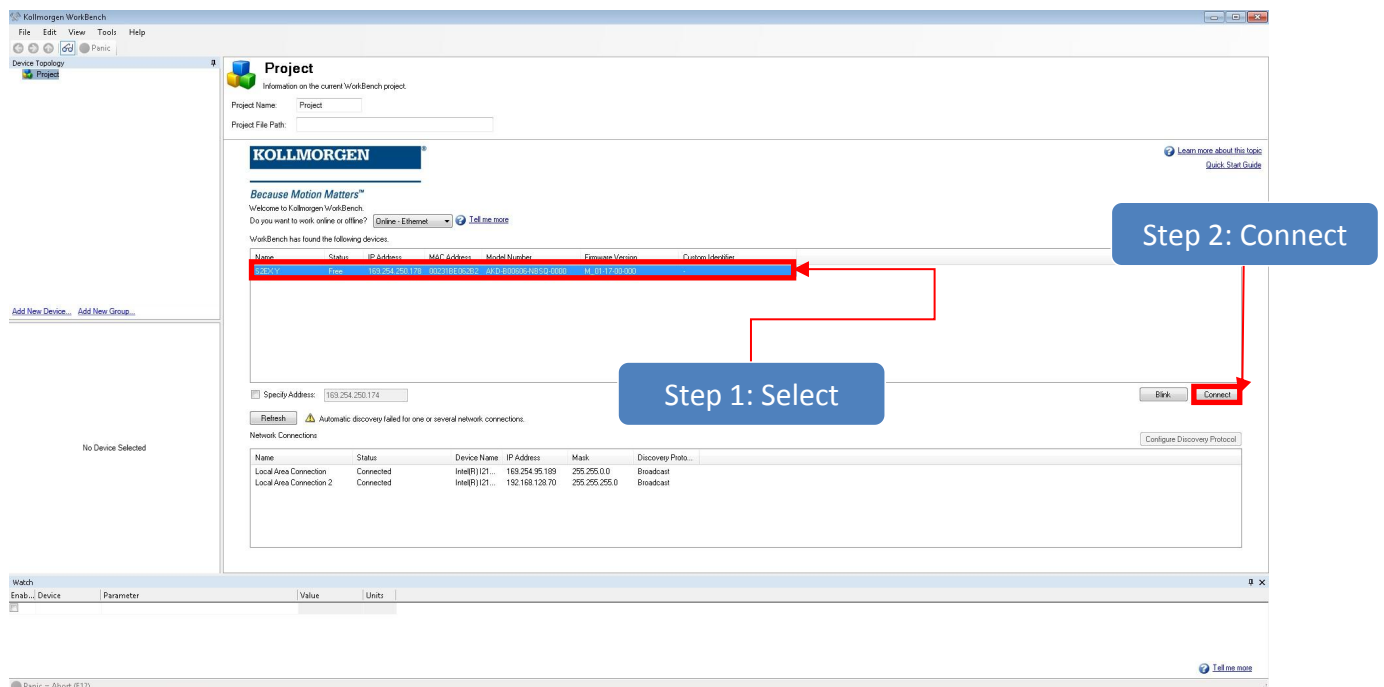


Figure 19: Log in the connected drive

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Upgrade Y-Axis Kollmorgen Drive Firmware Update

24. Interface will illustrate as below, Rename the drive with “Y-Axis” click “More” continues with “Update”

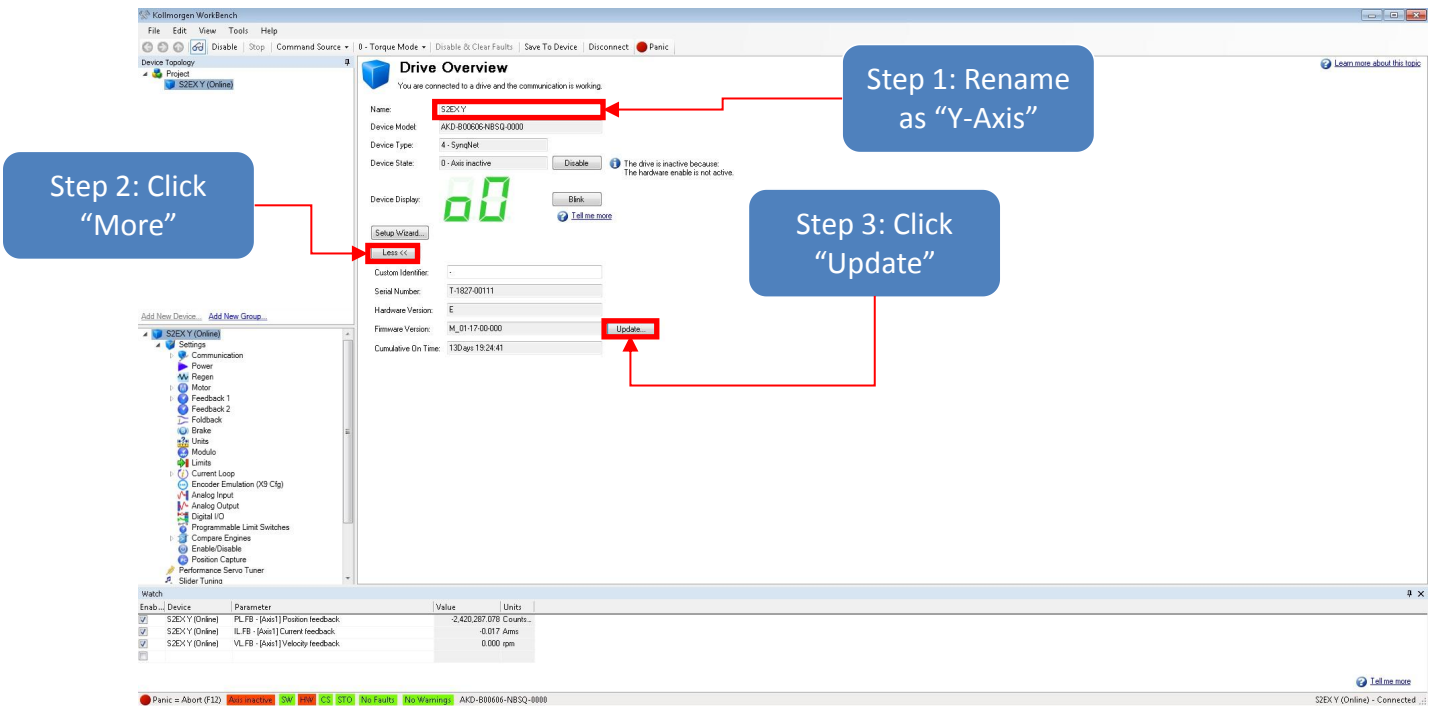


Figure 19: Log in the connected drive

25. Software will remind user that driver name has been changed and ask user whether to save this to device or not. Select “Yes” to continue.

26. Go to path “C:\Program Files\AXI System\v810\5.0\<Config>\motionControl\motionDrive\synqNet\AXI AKD Tuning Files” and select the “AKD-B-NBSQ-V01-17-00-000.i00”. Click “Open”.

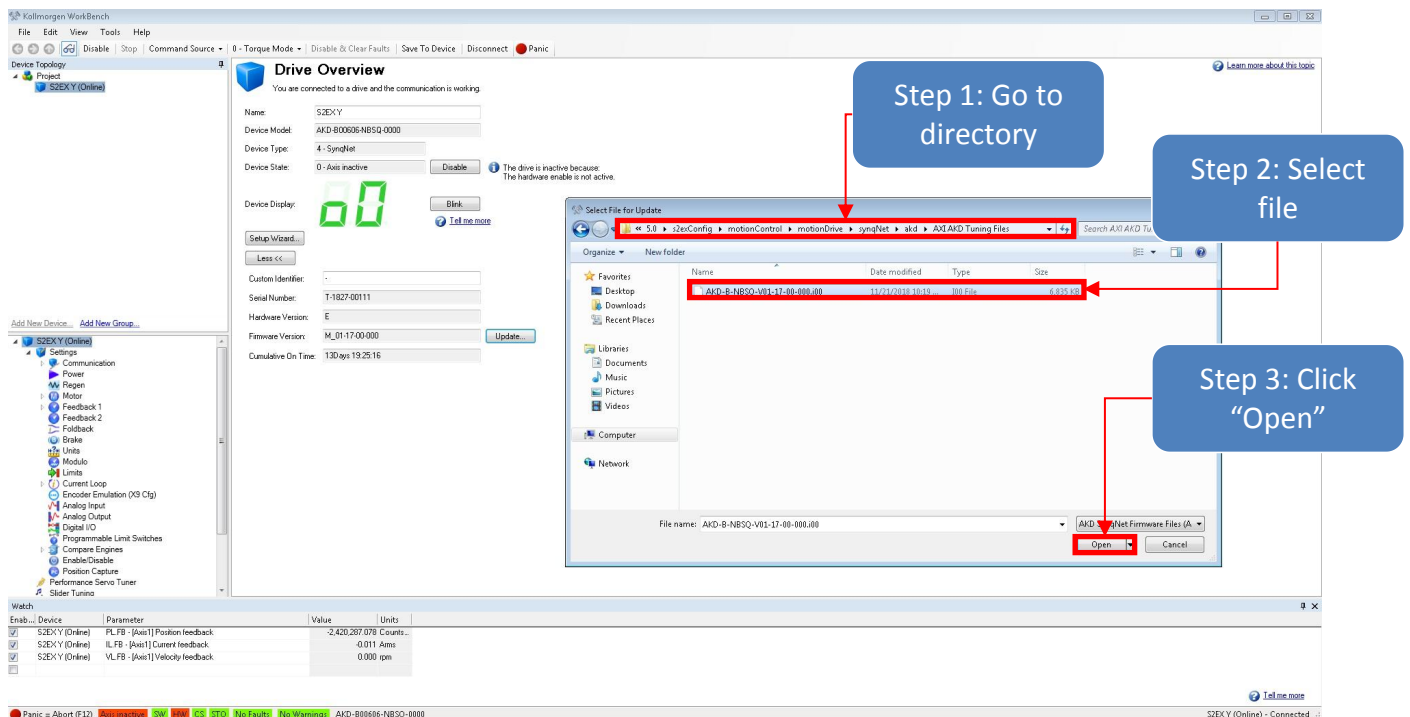


Figure 20: Select Firmware file

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27. The software will appear as loading status. It usually takes few minutes to load, please wait until it finish and the driver status pane will appear again. This means the firmware is loaded successfully.

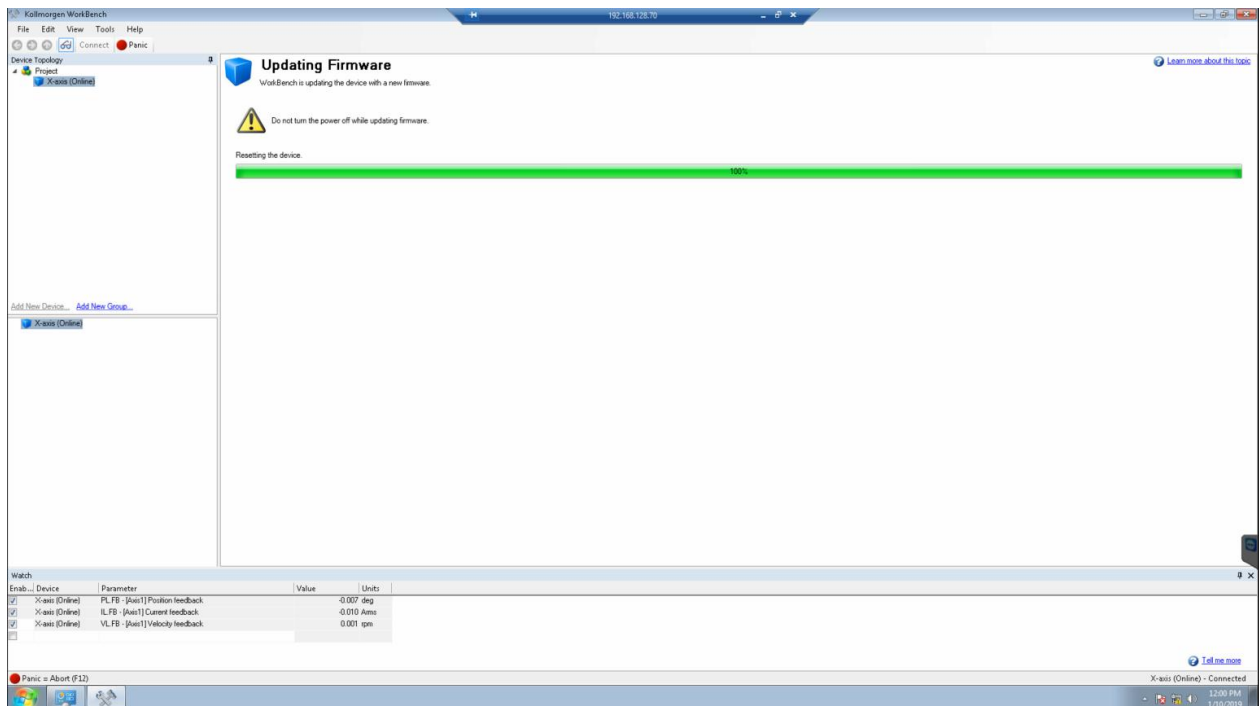


Figure 21: Complete update Firmware

Upgrade Y-Axis Kollmorgen Drive Parameter Update

28. At the same Kollmorgen WorkBench” application, click the “Tools” > “Y Axis Online” > “Load Parameter”.

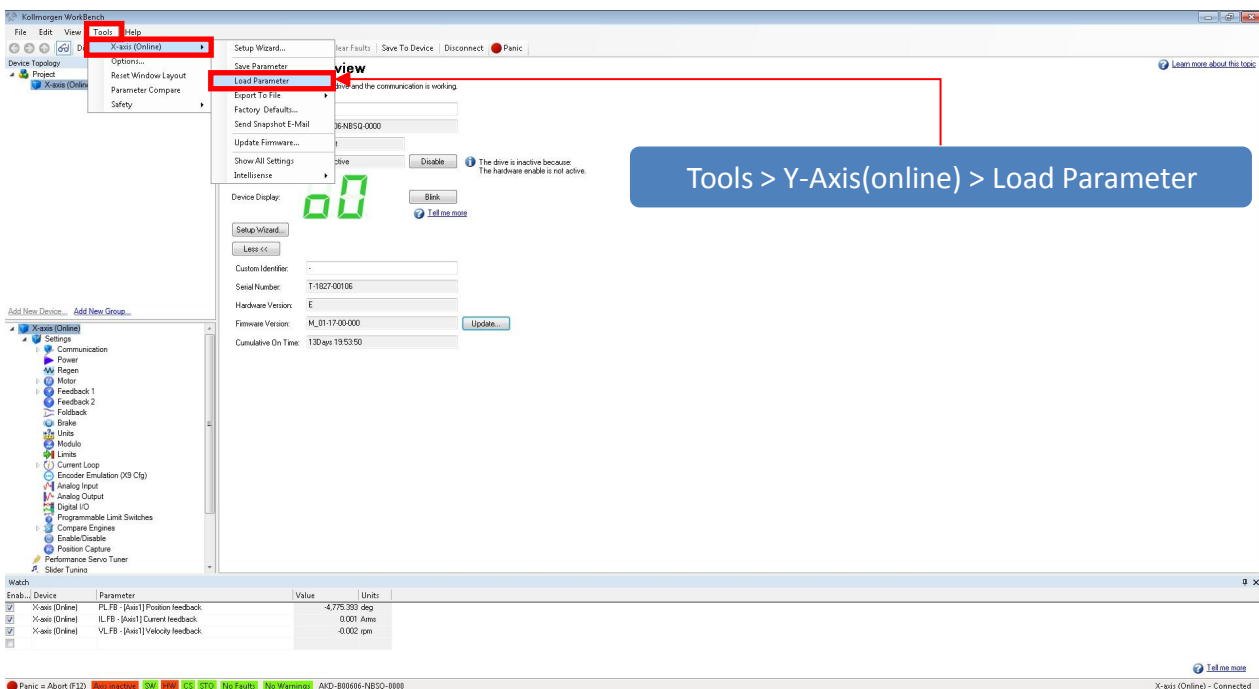


Figure 22: Go to “Load Parameter” function

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29. Go to path “C:\Program Files\AXI System\v810\5.0\<Config>\motionControl\motionDrive\synqNet\AXI AKD Tuning Files” and select the “y_axis.akd”. Click “Open”.

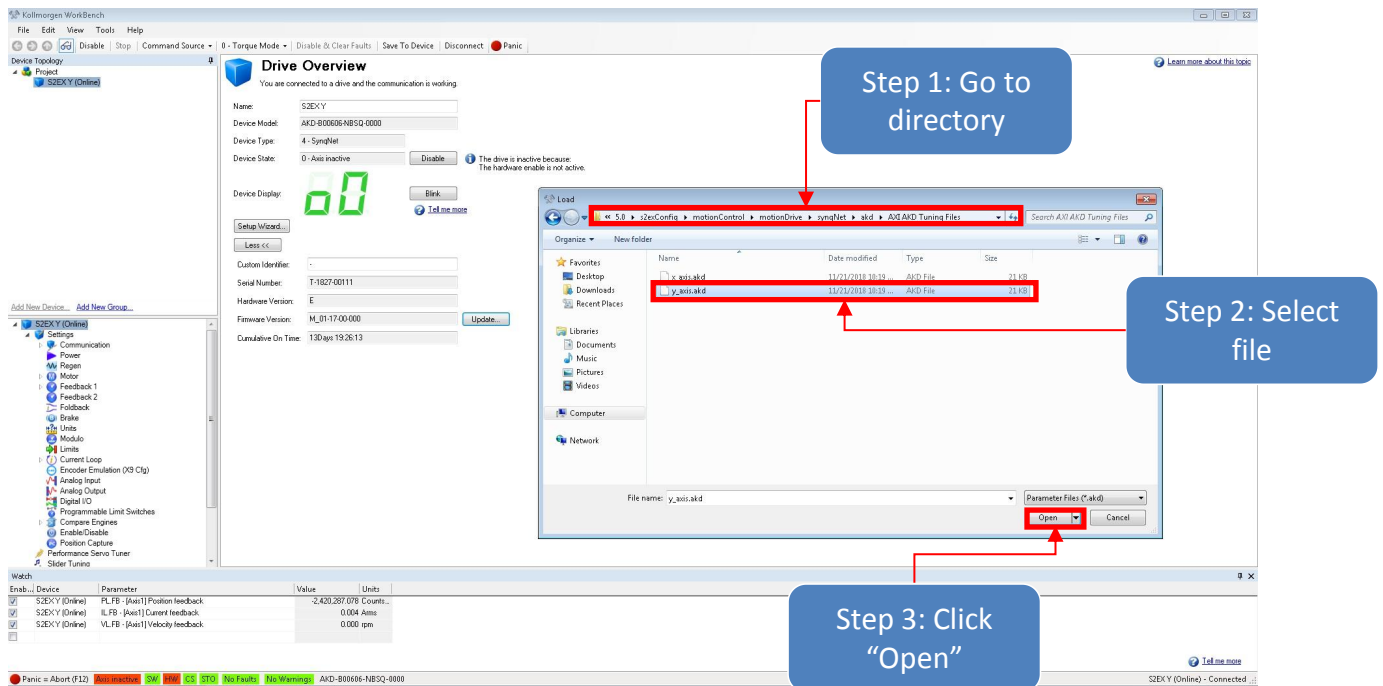


Figure 23: Select Y Axis Parameter file

30. A loading dialog will prompt out. When the parameter loaded successfully, a smiley face will show up. Then click “Save To Device” button and follow with “OK”.

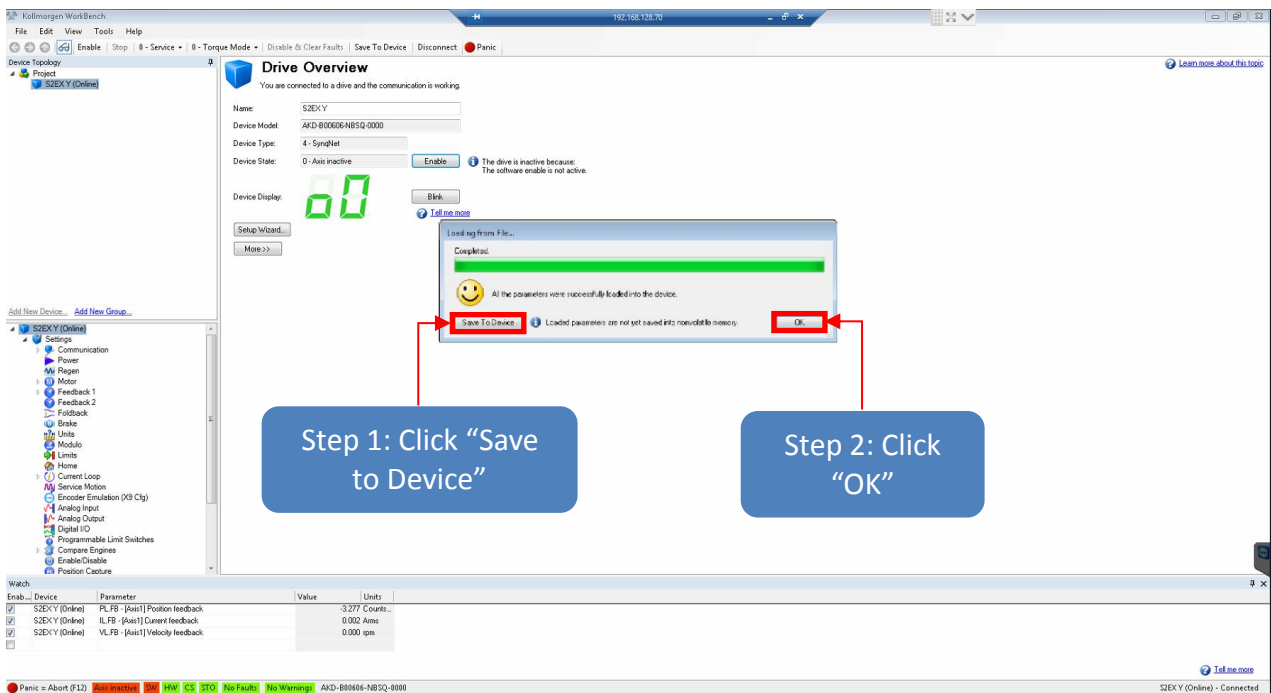


Figure 24: Save Parameter setting to X axis driver

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31. Once complete, exit the “Kollmorgen WorkBench” and click “No”. Close all the window.

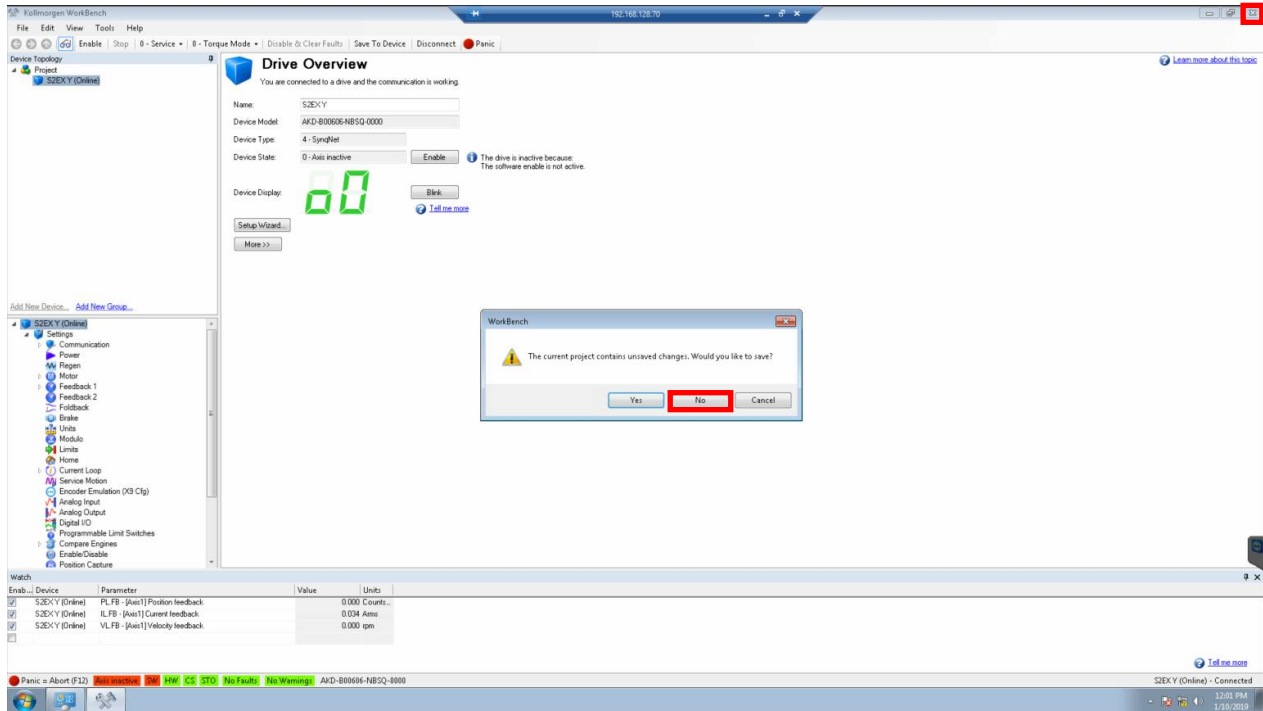


Figure 25: Save Parameter setting to X axis driver

32. Disconnect and remove the external LAN cable from Atom PC and AKD Driver.

33. Return to V810 GUI, go to Connected Processor tab with V810 GUI, click the "Stop RMS" > “Upgrade RMS” > “Restart RMS”# **MUST perform**

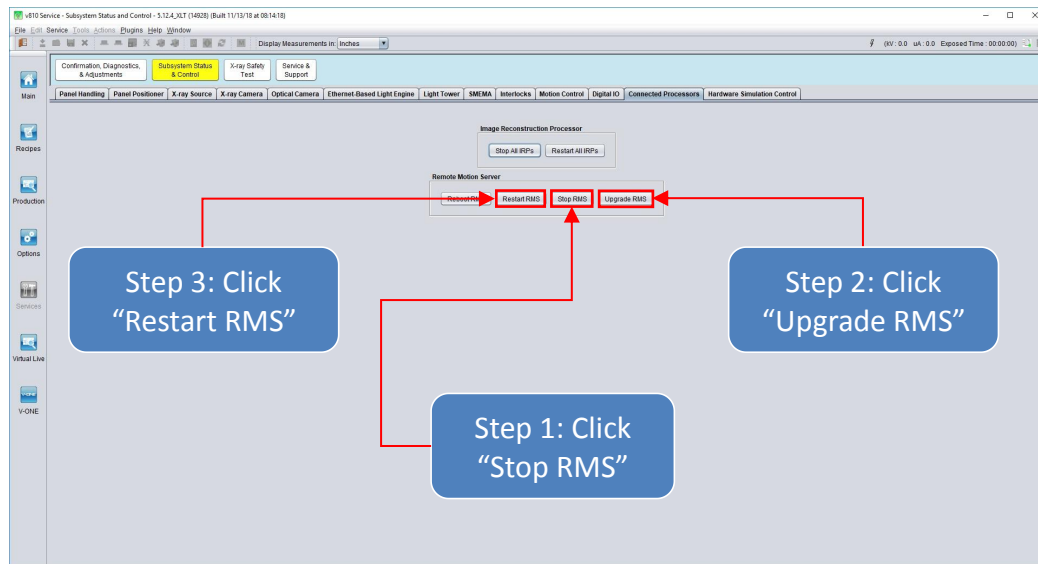


Figure 26: upgrade and restart RMS

34. Go to Motion Control Tab and click “ Upgrade Firmware”.

35. Run CD&A